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Task Description:

Multiphase flow separation is a critical technological challenge for the human exploration of space and planetary bodies. While droplet and bubble segregation is a spontaneously occurring phenomenon in most terrestrial situations, the absence of the gravity-induced separatory action in a microgravity environment often results in situations where two disparate phases have no distinct inclination to separate from one another. This can present many problems with respect to spacecraft and space station operations as proper control and sequestration of the individual phases is often essential to the successful functionality of many system-critical components.
Passive cyclonic separators are a promising technology that potentially could fulfill the separation needs of many of these systems. While considerable research has been performed on this technology, the limitations of terrestrial testing, aircraft testing, and modeling necessitates long-term experimentation to raise these devices to a sufficient technology readiness level (TRL) to permit use within spacecraft systems. It is a lack of basic engineering and scientific knowledge that currently prevents these systems from being employed. At a basic engineering level, there is a distinct lack of data regarding both the steady and unsteady operational performance of vortex separators.

Issues such as where the designs can be expected to work and how to go about optimizing designs are unresolved. Accurate prediction of these phenomena using computational fluid dynamics (CFD) is currently not tenable due to the dearth of available data and incomplete modeling techniques. Beyond engineering, there is also a lack of knowledge regarding the fundamental multiphase fluid physics phenomena that occur within passive cyclonic separators. Issues such as gas core adjustment to injection flow rates, bubble dynamics, vortex flow interaction with capillary forces at the gas core, etc. are all unresolved. Inasmuch, this research effort seeks to resolve these issues. The main objectives are to:

1. Quantitatively assess and delimit the range of operability of a prototype passive cyclonic separator including the transient and steady-state instability behavior of the device with respect to both device and system-level conduct.

2. Elucidate and quantify the multiphase fluids physics phenomena governing the performance of microgravity two-phase separators.

3. Validate and verify the multiphase CFD modeling approaches that have been developed to simulate the complicated two-phase transport processes associated with phase separation in microgravity.

4. Validate and verify a hydraulic system code for the simulation of two-phase flow in microgravity.

To achieve these objectives, combined experimental, numerical, and phenomenological modeling analyses are being performed to quantitatively describe the performance and response of systems containing separators. Specifically, the work entails the creation of multiphase flow system that is able to provide metered quantities of gas and liquid into a test separator. The separator and the associated flow loop had been operated under a variety of specially tailored conditions, including specific combinations of parameters to examine flow features of interest including the steady-state performance margins of the separator, the transient response of the separator, vapor core instability and misting, system startup/shutdown, and system level performance and instabilities. The products of this research have included the flight experiment (including two-phase flow loop and passive cyclonic separator), validated and verified CFD modeling techniques, a microgravity hydraulic system code, and archival data capturing the important fluid physics associated with microgravity separator operations.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

From a microgravity/scientific standpoint, the experimentation, modeling, and analytical work of this research will serve to:
1. Provide elucidation and understanding of the above-mentioned two-phase flow phenomena associated with the operation of separators in multiphase systems

2. Provide details of the columnar stability of the gas core

3. Provide details of the time constants associated with the response of passive cyclonic separators

4. Validate and verify a coupled CFD and hydraulic system code for simulations of two-phase flow in microgravity

The resulting insight into phase separation, distribution, and control will help to afford NASA designers the latitude to take greater advantage of the benefits offered by the use of these multiphase systems in spacecraft applications. The experimental results will also facilitate further new and original design modifications to improve the performance of microgravity phase separation devices. Validation and verification of the numerical results are expected to permit the use of CFD as a reliable design tool to further assist in the creation and analysis of new separator designs, while the results from the system code are expected to assist in the creation and analysis of general multiphase systems in microgravity.

Overall, the research will help contribute to the fundamental scientific understanding of two-phase fluid behavior, both in microgravity and 1-g environments.

Numerical Modeling
In the current reporting period, our focus was on the effects of surface tension, or more specifically, the collapse of the gas-core due to surface tension. It is important to maintain a stable gas core in order for the separator to perform properly. Therefore, it is necessary to know the conditions under which a stable gas core exists. Although the hydrodynamic stability of rotating liquid layer(s) have been investigated in the past, the present study deals with a moving boundary problem in which the boundary collapses due to unbalanced forces, not due to amplifications of small disturbances. Although our eventual objective is to study the performance of cyclone separator in microgravity, the present study focuses on liquid-only injection to obtain the fundamental information regarding the stability of gas-core.

One important aspect of the simulation is the existence of the liquid-gas interface. A well-known interface tracking scheme called the volume-of-fluid (VOF) method is employed in the present analysis. Both fluids are assumed to be incompressible. Only one set of volume fraction transport and momentum equations are solved for the two fluids. Surface tension is included as a source term in the momentum equation. Two fluids are distinguished by a scalar variable called the volume fraction, which is computed by solving a transport equation. The interface location is then identified as the location where the volume fraction changes sharply. The no-slip condition is applied on all solid

Task Progress:	surfaces. The contact angle is assumed to be 90 degrees. From the simulations it is found that if the Weber number (We) is reduced from that of a stable gas core, the gas core will eventually collapse below a certain We. The collapsing of the gas core starts near the bottom baffle wall. The interfacial Weber number is proportional to the gas core radius and the square of the tangential velocity at the interface, so it is sensitive to the tangential velocity variation in the axial direction. The tangential velocity goes to zero at the baffle plate due to the no-slip condition. As a result, the surface tension effects are largest near the baffle plate so that the gas core collapses starting from this region. The values of We0, min (minimum critical Weber number for collapsing) are also investigated in the parametric study. Analytical Modeling In the previous work (Hoyt et al., 2013), steady-state performance of liquid-only injection was studied. For liquid-only injection, the previously derived control-volume equations showed that the film thickness will only change if C_f changes. However, our later experiments showed that under the same liquid flow rate, the liquid film thickness changes with different injection nozzle diameter, which is related to parameter G in the control volume equations. Therefore, the correlation of C_f with the Reynolds number and G was investigated. Then, a new set of control volume equations were derived and solved. The results were compared with our experiment data, which showed a good agreement. Experimentation In our original science requirement document, it is specified that the development pipe shall have a minimum linear distance of 25" (63.5 cm) for the flow development between the two-phase mixer and the test section. The development pipe shall be as long as possible so that the two-phase flow will fully develop. However, in order to satisfy the engineering requirements, the development pipe needs to be short enough to fit into the test rig. Therefore, tests were conducted to investigate the influence of different development pipe lengths on the bubble size distribution. Pipes of three different lengths, 2’’, 12’’, and 25’’, were used for this experiment. A high-speed camera with a frame rate of 1000 fps was used to capture the pictures of bubble distribution in the water tank. The high-speed images were further analyzed with ImageJ software. It is found that the typical bubble diameter is on the order of mm. The average bubble size becomes smaller with increasing liquid flow rate. The effect of gas flow rate is related to bubble coalescence, which creates large bubbles. With increasing gas volume fraction, more bubbles will coalesce in a given time. On the other hand, for a given liquid flow rate, the residence time in the pipe decreases with increasing gas volume fraction, which decreases time for coalescence. The data suggests that more large bubbles (>2 mm) appear with decreasing gas volume fraction, which indicates the importance of residence time in the pipe. Supporting Work for Flight Hardware Design A breadboard model of the proposed experiment is being designed and built at NASA/Glenn Research Center (GRC). We have collaborated with NASA/GRC and ZIN engineers to prepare and design the breadboard experiment system in past year. Planned Work and Facility Utilization Extensive collaboration with NASA personnel and utilization of NASA facilities is to continue in the upcoming year. Preparation for the eventual flight experiment will be continued in the coming year in concert with the above studies. The computational work will be continued at Case Western Reserve University (CWRU) in the upcoming year with the objective of augmenting the maturity of CFD as a design tool for this class of fluid flows. Reference Hoyt NC, Kang MF, Lee KL, Kharraz A, Kadambi J, Kamotani Y. "Study of steady and dynamic behavior of gas core of passive cyclonic separator for space applications." Microgravity Science and Technology. 2013 Oct;25(3):187-200.
Bibliography Type:	Description: (Last Updated: 03/06/2024)
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Abstracts for Journals and Proceedings	Kamotani Y. "Investigation of Gas-Liquid Separator Performance in Microgravity Environment." 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Dissertations and Theses	Kharraz A. (Adel Kharraz) "Stability of Swirling Flow in Passive Cyclonic Separator in Microgravity." Dissertation, Case Western Reserve University, Cleveland, OH, January 2018. , Jan-2018